IMO(49th) – 2008

First Day

1. An acute angled triangle ABC has orthocenter H. The circle passing through H with centre the midpoint of BC intersects the line BC at A_1 and A_2. Similarly, the circle passing through H with centre the midpoint of CA intersects the line CA at B_1 and B_2, and the circle passing through H with centre the midpoint of AB intersects the line AB at C_1 and C_2 show that $A_1, A_2, B_1, B_2, C_1, C_2$ lies on a circle.
2.
   (a) Prove that $\frac{x^2}{(x-1)^2}+\frac{y^2}{(y-1)^2}+\frac{z^2}{(z-1)^2}\geq 1$ for all real numbers x, y, z each different from 1 and satisfying $xyz=1$.
   (b) Prove that equality holds above for infinitely many triples of rational numbers x, y, z, each different from 1 and satisfying $xyz=1$.
3. Prove that there exist infinitely many positive integers n such that n^2+1 has a prime divisor which is greater than $2n+\sqrt{2n}$.

Second Day

4. Find all functions $f:(0,\infty)\to(0,\infty)$ (so, f is a function from positive real numbers to the positive real numbers) such that $\frac{(f(w))^2+(f(x))^2}{(f(y))^2+(f(z))^2}=\frac{w^2+x^2}{y^2+z^2}$ for all real numbers w, x, y, z satisfying $wx=yz$.
5. Let n and k be positive integers with $k\geq n$ and $k-n$ an even numbers. Let $2n$ lamps labeled $1,2,...,2n$ be given, each of which can be either on or off. Initially all the lamps are off. We consider sequences of steps: at each step one of the lamps is switched (from on to off or from off to on).
   Let N be the number of such sequences consisting of k steps and resulting in the state where lamps 1 through n are all on, and lamps $n+1$ through $2n$ are all off.
   Let M be the number of such sequences consisting of k steps and resulting in the state where lamps 1 through n are all on, and lamps $n+1$ through $2n$ are all off, but where none of the lamps $n+1$ through $2n$ is ever switched on.
   Determine N/M.
6. Let ABCD be a convex quadrilateral with $|BA|\neq|BC|$. Denote the incircles of triangles ABC and ADC by w_1 and w_2 respectively. Suppose that there exist a circle w tangent to the ray BA beyond A and to the ray BC beyond C, which is also tangent to the lines AD and CD. Prove that the common external tangents w_1 and w_2 intersect on w

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